(部分：10.26 最后 Proof)

Theorem : $T: V \to V$ is Δ-able

$\Leftrightarrow$

$\chi_T(x) = (x-\lambda_1) \cdots (x-\lambda_n)$

Corollary : V is a $\mathbb{C}$-v. space (complex)

$\Rightarrow \forall T: V \to V$ is Δ-able

Proof : By fundamental thm. of algebra,

$\forall \chi_T(x)$ can be factorized linearly

Theorem 1 : $T: V \to V$ is Δ-able

$\Rightarrow$

$\chi_T(T) = \underline{0}_{V \to V}$

Corollary 2 : (C.-H. for $\mathbb{C}$-v. spaces $\overset{(T: V \to V \text{ for})}{=\!=}$)

$\forall T: V \to V$, where V is $\mathbb{C}$-v. space

$\Rightarrow \chi_T(T) = \underline{0}_{V \to V}$

$\Rightarrow m_i(x) \mid \chi_T(x)$

Proof (of ——②)

Suppose $\exists \mathcal{B}$ a basis ——②(of $V_{n\times 1}$)

s.t. $T_{\mathcal{B}\mathcal{B}} = \begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$

Consider $\chi_T(x) = (x-\lambda_1)\cdots(x-\lambda_n)$

(and study $\chi_T(T) = (T-\lambda_1 I)\cdots(T-\lambda_n I) : V \to V$

(and check $\underline{0}_{V\to V}$))

$\Rightarrow \chi_T(T)_{\mathcal{B}\mathcal{B}} = (T_{\mathcal{B}\mathcal{B}} - \lambda_1 I_{\mathcal{B}\mathcal{B}}) \cdots (T_{\mathcal{B}\mathcal{B}} - \lambda_n I_{\mathcal{B}\mathcal{B}}) \in \underbrace{n\times n \text{ matrix}}$

(and check $\chi_T(T)_{\mathcal{B}\mathcal{B}} = 0_{n\times n}$)

$= \begin{pmatrix} 0 & & * \\ & \lambda_2-\lambda_1 & \\ & & \ddots \\ 0 & & \lambda_n-\lambda_1 \end{pmatrix} \cdots \begin{pmatrix} \lambda_1-\lambda_n & & * \\ & \ddots & \\ & & \lambda_{n-1}-\lambda_n \\ 0 & & 0 \end{pmatrix}$

$= A_1 \qquad\qquad = A_n$

Claim: $\forall \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{F}^n$, $\underbrace{(A_1 \cdots A_n)}_{=0_{n\times n}} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \underline{0}_{\#}$ ——③

Proof of Claim:

$A_1 \cdots A_{n-1} \cdot A_n \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

$= A_1 \cdots A_{n-1} \begin{pmatrix} x_1' \\ \vdots \\ x_{n-1}' \end{pmatrix}$

$= A_1 \cdots A_{n-2} \begin{pmatrix} x_1'' \\ \vdots \\ x_{n-2}'' \end{pmatrix}$

$= \cdots$

Corollary (C.-H. for $\mathbb{R}$-v. spaces):

Let $T: V \to V$ be any L. operators,
where V is a $\mathbb{R}$-v. space

then $\chi_T(T) = \underline{0}_{V \to V}$

$\Rightarrow m_T(x) \mid \chi_T(x)$

(Note: not all $T: V \to V$ (V are $\mathbb{R}$-space) are Δ-able, so Thm. 1 doesn't apply)

Proof:

Let $\mathcal{B}$ be any basis (real) of V

Consider $M := T_{\mathcal{B}\mathcal{B}} \in M_{n\times n}(\mathbb{R})$ (①, 2040)

Then $\chi_T(x) = \chi_M(x)$ (①, 2040)

(② defined in MAT2040)

Consider $M: \mathbb{C}^n \to \mathbb{C}^n$

$\underline{x} \to M\underline{x}$ (2040)

($\mathbb{C}^n$ use usual basis $\mathcal{C} = \{\underline{e}_1, \cdots, \underline{e}_n\}$,

$M_{\mathcal{C}\mathcal{C}} = M$) (2040)

① $\Rightarrow \chi_M(x) = \chi_M(x)$

(?) (3040) (2040)

Apply Corollary 1 (C.-H. for $\mathbb{C}$-v. space) on M:

$\chi_M(M) = 0_{\mathbb{C}^n \to \mathbb{C}^n} \Rightarrow \chi_M(M) = 0_{\mathbb{C}^n \to \mathbb{C}^n}$ (2040)

$\Rightarrow$ —— (見證)

Conclusion: $m_T(x) \mid \chi_T(x)$

Theorem 3 :

Let $T: V \to V$, V: $\mathbb{C}$- v. space

with $\chi_T(x) = (x-\lambda_1)^{e_1} \cdots (x-\lambda_k)^{e_k}$

call: irreducible $p_i(x) \in \mathbb{C}[x]$ are $p_i(x) = x - \lambda_i$ linear

Then $m_T(x) = (x-\lambda_1)^{f_1} \cdots (x-\lambda_k)^{f_k}$

with $1 \le f_i \le e_i$, $\forall i$

Proof:

Consider $\chi_T(x) = (x-\alpha)^{e_1} (x-\beta)^{e_2}$

Suppose on contrary $m_T(x) = 1 \cdot (x-\beta)^{f_2}$

Then (by $\chi_T(x)$) $\exists$ e.vector $\underline{v} \neq \underline{0}$

with $T(\underline{v}) = \alpha \underline{v}$ ——(1)

(by def. of $m_T(x)$) $(T-\beta I)^{f_2} = \underline{0}_{V \to V}$ ——(2)

So

(2) $\Rightarrow$ $\underline{0}_{V \to V}(\underline{v}) = (T-\beta I)^{f_2}(\underline{v})$

$\Rightarrow \underline{0}_V = \underbrace{(T-\beta I) \cdots (T-\beta I)}_{f_2}(\underline{v})$

$= (T-\beta I) \cdots (T-\beta I)(T\underline{v} - \beta \underline{v})$

$\Rightarrow$